MATH 8090: State-Space Models I

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Local Level Model: Simulation and Signal-to-Noise Ratio

Consider the local-level model

$$X_t = X_{t-1} + V_t, \qquad Y_t = X_t + W_t,$$

where V_t and W_t are independent Gaussian white-noise processes.

The signal-to-noise ratio is defined as

$$\text{SNR} = \frac{\sigma_V^2}{\sigma_W^2}.$$

Comparing Different Signal-to-Noise Ratios

We construct two observed series using the same latent state and the same standardized observation noise.

For a fixed state innovation variance, a larger SNR corresponds to smaller measurement noise variance. Consequently, the observed series should track the latent state more closely.

```
set.seed(123)

# Initial state: X_1 ~ N(m.1, sig.1)
m.1 <- 0
sig.1 <- 1
```

```r
# State innovation variance
sig2.V <- 1

# Simulate the latent random walk
X.1 <- rnorm(1, mean = m.1, sd = sqrt(sig.1))
X <- cumsum(c(X.1, rnorm(99, sd = sqrt(sig2.V))))

# Common observation noise, rescaled for each SNR
W <- rnorm(100)
```

Comparing Different Signal-to-Noise Ratios

We construct two observed series using the same latent state and the same standardized observation noise. For a fixed state innovation variance, a larger SNR corresponds to smaller measurement noise variance. Consequently, the observed series should track the latent state more closely.

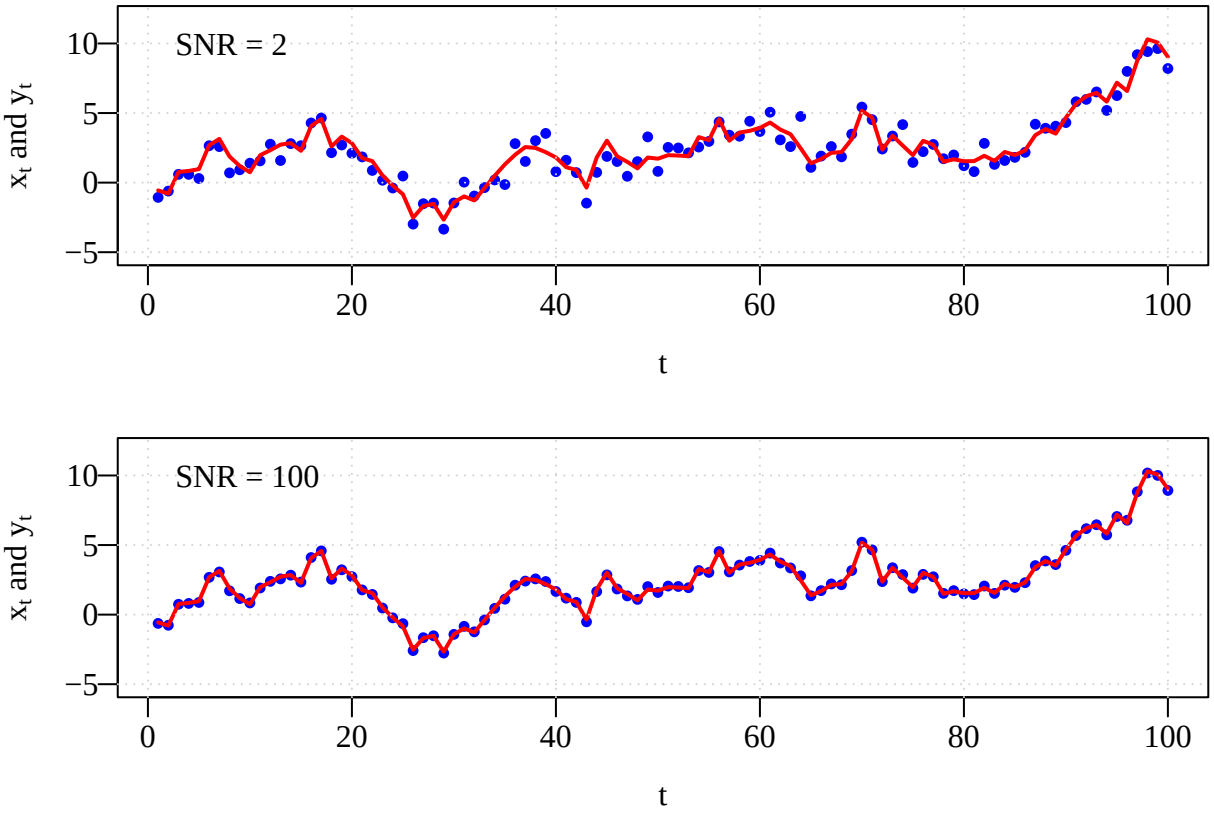
```r
# SNR = 2: relatively noisy observations
SNR <- 2
Y.2 <- X + W * sqrt(sig2.V / SNR)

# SNR = 100: much smaller measurement noise
SNR <- 100
Y.100 <- X + W * sqrt(sig2.V / SNR)

# Compare the two observed series
par(las = 1, mgp = c(2, 0.5, 0),
    mar = c(3.5, 3.5, 1, 0.6), mfrow = c(2, 1), family = "serif")

plot(Y.2, col = "blue", pch = 16, cex = 0.75, xlab = "t",
     ylab = expression(paste(x[t], " and ", y[t])), main = "", ylim = c(-5.25, 12))
lines(X, col = "red", lwd = 2)
grid()
legend("topleft", legend = "SNR = 2", bty = "n")

plot(Y.100, col = "blue", pch = 16, cex = 0.75, xlab = "t",
     ylab = expression(paste(x[t], " and ", y[t])), main = "", ylim = c(-5.25, 12))
lines(X, col = "red", lwd = 2)
grid()
legend("topleft", legend = "SNR = 100", bty = "n")
```



Interpretation: The red line represents the latent state, while the blue points represent noisy observations. When SNR increases from 2 to 100, the observations become more concentrated around the true state.

This motivates the filtering problem: how can we recover the latent state when observations are contaminated by noise?

Prediction, Filtering and Smoothing for the Local Level Model

This example follows Example 6.5 of Shumway and Stoffer (2017).

We compare three approaches to estimating the latent state: - Forecasting: Estimate X_t using observations through time $t-1$. - Filtering: Estimate X_t using observations through time t. - Smoothing: Estimate X_t using the complete observed series.

Generate Data

We simulate a local-level model with unit state innovation variance and unit observation noise variance.

```
set.seed(1)

n <- 50

# Initial state X_0 and subsequent state innovations
w <- rnorm(n + 1, 0, 1)

# Independent observation noise
v <- rnorm(n, 0, 1)
```

```r
# mu[1] = X_0; mu[-1] = X_1, ..., X_n
mu <- cumsum(w)

# Observations Y_t = X_t + W_t, t = 1, ..., n
y <- mu[-1] + v
```

Filtering and Smoothing

We use `Ksmooth()` from the **astsa** package to compute the Kalman predictions, filtered estimates, and smoothed estimates.

The model parameters are treated as known, allowing us to focus on state estimation rather than parameter estimation.

```r
library(astsa)

# Initial-state distribution
mu0 <- 0
sigma0 <- 1

# State transition and noise variances
phi <- 1
sQ <- 1  # State innovation variance
sR <- 1  # Observation noise variance

# Kalman filtering and smoothing
ks <- Ksmooth(y, A = 1, mu0 = mu0, Sigma0 = sigma0, Phi = phi, sQ = sQ, sR = sR)
```

Compare Forecasting, Filtering, and Smoothing

The following plots compare the true latent state with each estimate.

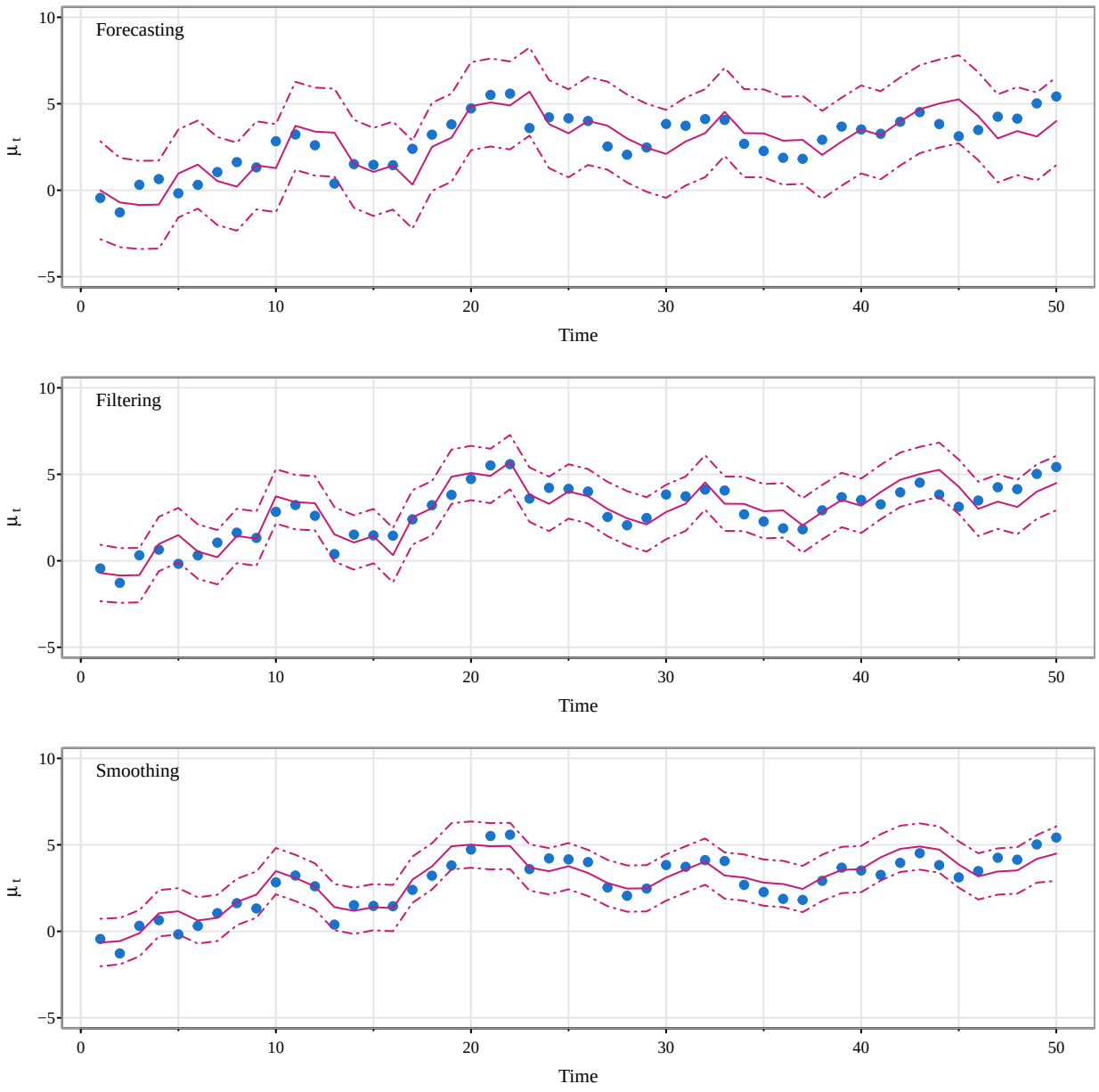
The dashed curves represent the estimate plus or minus two estimated standard deviations. Under the Gaussian model, these correspond to approximately 95% pointwise uncertainty intervals.

```r
par(mfrow = c(3, 1), mar = c(3.5, 3.5, 0.5, 0.5),
    mgp = c(2, 0.5, 0), family = "serif", las = 1)

# 1. One-step-ahead prediction
tsplot(mu[-1], type = "p", col = 4, pch = 19, ylab = expression(mu[~t]), ylim = c(-5, 10))
lines(ks$Xp, col = 6)
lines(ks$Xp + 2 * sqrt(ks$Pp), lty = 6, col = 6)
lines(ks$Xp - 2 * sqrt(ks$Pp), lty = 6, col = 6)
legend("topleft", legend = "Forecasting", bty = "n")

# 2. Filtering: uses observations through time t
tsplot(mu[-1], type = "p", col = 4, pch = 19, ylab = expression(mu[~t]), ylim = c(-5, 10))
lines(ks$Xf, col = 6)
lines(ks$Xf + 2 * sqrt(ks$Pf), lty = 6, col = 6)
lines(ks$Xf - 2 * sqrt(ks$Pf), lty = 6, col = 6)
legend("topleft", legend = "Filtering", bty = "n")
```

```r
# 3. Smoothing: uses the complete observed series
tsplot(mu[-1], type = "p", col = 4, pch = 19, ylab = expression(mu[~t]), ylim = c(-5, 10))
lines(ks$Xs, col = 6)
lines(ks$Xs + 2 * sqrt(ks$Ps), lty = 6, col = 6)
lines(ks$Xs - 2 * sqrt(ks$Ps), lty = 6, col = 6)
legend("topleft", legend = "Smoothing", bty = "n")
```




```r
# Average conditional variance of each estimator
c(Prediction = mean(ks$Pp), Filtering  = mean(ks$Pf), Smoothing  = mean(ks$Ps))
```

```
## Prediction  Filtering  Smoothing
##  1.6268090  0.6191697  0.4517972
```

Interpretation: Prediction uses the least information, filtering incorporates the current observation, and

smoothing also uses future observations. We therefore generally expect smoothing to provide more precise estimates of past states.

In particular, the conditional variances satisfy

$$\mathrm{Var}(X_t \mid Y_{1:n}) \le \mathrm{Var}(X_t \mid Y_{1:t}) \le \mathrm{Var}(X_t \mid Y_{1:t-1}), \qquad t \le n.$$

Initial State Estimate

Finally, compare the true initial state X_0 with its smoothed estimate and associated uncertainty.

```
# True initial state X_0
mu[1]
```

```
## [1] -0.6264538
```

```
# Smoothed estimate E(X_0 | Y_1, ..., Y_n)
ks$X0n
```

```
##             [,1]
## [1,] -0.3241541
```

```
# Posterior standard deviation of X_0
sqrt(ks$P0n)
```

```
##           [,1]
## [1,] 0.7861514
```

The Kalman smoother uses all available observations to estimate the initial state, even though X_0 is never observed directly.

References

Shumway, Robert H, and David S Stoffer. 2017. *Time Series Analysis and Its Applications*. 4th ed. Springer.